RecallGPT

YOUR KNOWLEDGE. ORGANIZED. UNDERSTOOD. ALWAYS WITH YOU.

An AI-powered Personal Knowledge Management System that understands your notes, tasks, expenses and memories using Semantic Search, RAG and AI Automation.

- **SMART RETRIEVAL** — Find what matters instantly
- **AUTOMATED BOOKKEEPING** — Extract tasks & expenses effortlessly
- **PHOTO MEMORY** — Exif-aware photo search & management
- **AI CHAT ASSISTANT** — Ask anything, get intelligent answers

ABOUT RECALLGPT

RecallGPT helps you store, organize and retrieve your personal knowledge intelligently.

It converts unstructured notes into structured insights using Semantic Search, RAG (Retrieval Augmented Generation) and AI Automation.

Notes · Tasks · Expenses · Photos

KEY FEATURES

SEMANTIC KNOWLEDGE RETRIEVAL
Search your notes naturally. RecallGPT understands context and delivers the most relevant information.

AUTOMATED BOOKKEEPING
Automatically extracts Tasks and Expenses from notes using NLP and regex patterns.

EXIF-AWARE PHOTO MEMORY
Fetch your photos by location, date or context using metadata understanding.

AI CHAT ASSISTANT (RAG POWERED)
Ask questions. RecallGPT searches your knowledge base and answers using Llama 3.1 8B via NVIDIA API.

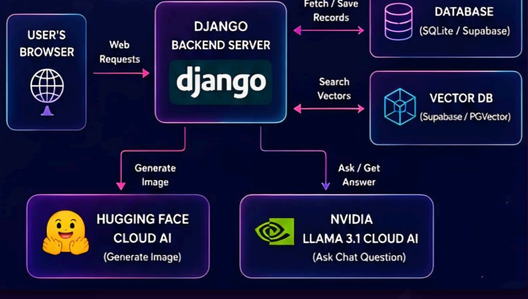
TECH STACK

- **Backend** — Django (Python)
- **Database** — SQLite / Supabase
- **Vector Storage** — PGVector (Supabase)
- **Embedding Model** — all-MiniLM-L6-v2
- **LLM (Chat)** — Meta Llama 3.1 8B (via NVIDIA API)
- **Image Generation** — Hugging Face Cloud (Flux Schnell / SDXL)
- **Other Tools** — Pillow (Exif), Regex, REST API

HOW DOES IT WORK ?

1. **INGEST** — Notes are split into chunks, embedded using all-MiniLM-L6-v2 and stored in PGVector.
2. **RETRIEVE** — User query is converted to vector and top-K relevant chunks are retrieved.
3. **AUGMENT** — Retrieved context is combined with query to create a rich prompt.
4. **GENERATE** — Prompt is sent to Llama 3.1 8B via NVIDIA API to generate accurate answers.
5. **RESPONSE** — Response is returned to the user with source-aware answers.

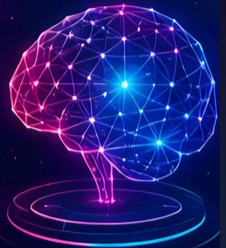
SYSTEM ARCHITECTURE



IMPACT

- Saves time by organizing personal knowledge intelligently.
- Reduces manual effort in tracking tasks and expenses.
- Helps recall memories through smart photo search.
- Provides accurate answers with source-aware responses.

RAG POWERED BY LLM



Open-book AI that combines the power of retrieval with generative intelligence to give accurate, context-aware and hallucination-reduced responses.

FUTURE SCOPE

- Offline support and local LLM deployment (Ollama)
- Mobile application for on-the-go access
- Advanced multimodal models (Llama-3.2-Vision, Qwen-VL)
- Personalized insights and proactive suggestions
- User studies and privacy-focused enhancements

EVALUATION & RESULTS

RETRIEVAL PERFORMANCE

Metric	Value
Recall@1	0.79
Recall@3	0.90
Recall@5	0.92
Mean Cosine Similarity	0.86
Retrieval Latency (Avg.)	34.8 ms

EXTRACTION PERFORMANCE

Metric	Value
Precision	91.8%
Recall	87.6%
F1-Score	89.6%
Extraction Accuracy	90.4%

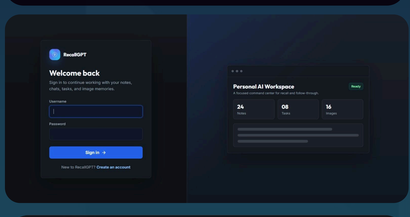
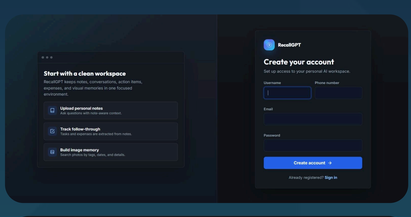
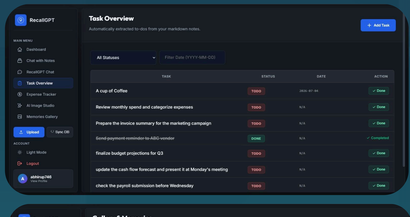
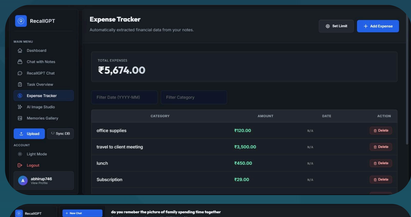
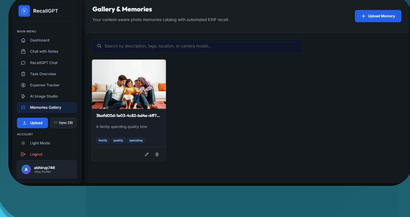
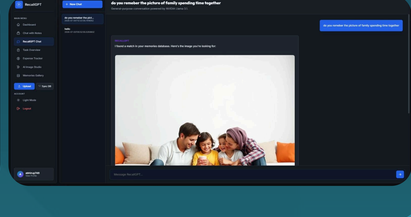
ERROR ANALYSIS (Total Errors = 66)

Error Type	Count	%
Informal writing style	15	22.7%
Abbreviated expressions	10	15.2%
Multiple expenses in one sentence	8	12.1%
Complex task descriptions	12	18.2%
Ambiguous date expressions	6	9.1%
Incomplete financial information	15	22.7%
Total	**66**	**100%**

USER SATISFACTION (Survey)

4.6 / 5

Average Rating

Based on a survey of 35 test users

IMAGES OF THE WORKING MODEL